

CAA Compliance Pipeline Data Dictionary

File: PIPELINE_CAA_00_COMPLETE.csv. Source: EPA ECHO, CAA Pipeline Download Summary, <https://echo.epa.gov/tools/data-downloads/caa-pipeline-download-summary>.

What it is. Per EPA, the pipeline “shows links between Compliance Monitoring Activities (CMA) to any related violations and/or enforcement actions” — i.e. it pre-joins, in a single row, a violation to its triggering evaluation and resulting enforcement action, a linkage not available in the individual ICIS-Air tables. The narrative structure and caveats are documented separately in `pipeline_brief.tex`; this file documents the 35 columns.

Conventions used below. “*EPA*” marks a definition quoted from EPA’s pipeline documentation; “*data*” marks a fact verified directly against the file. The file has 66,655 rows; FOUND_VIOLATION = Y for every row (*data*).

Identification and row-level flags

Field	Description
SORT_ORDER	EPA internal ordering of pipeline rows
SORT_DATE	Date used for EPA’s internal ordering
SOURCE_ID	Facility identifier (ICIS-Air PGM_SYS_ID)
REGISTRY_ID	Facility Registry Service (FRS) ID
AIR_NAME	Facility name
PIPELINE_FLAG	<i>EPA</i> : “an internally generated flag to indicate at least [one] violation (VIOL_ACTIVITY_ID) linked to an evaluation and/or enforcement action.” <i>Data</i> : values Y/N
OFFICIAL_FLAG	<i>EPA</i> : “this flag is set to ‘Y’ when an entry is counted as compliance monitoring strategy activities.”

Evaluation block (the compliance-monitoring activity)

Field	Description
EVAL_FLAG	<i>EPA</i> : “an internally generated flag to indicate if there is at least one Compliance Monitoring Activity (CMA) (evaluation) [linked] to the violation activity id.”
EVAL_SORT_ORDER	Internal ordering of the evaluation within the row
EVAL_ACTIVITY_ID	Evaluation activity identifier. <i>Data</i> : a sentinel value -9999 is used where no evaluation is linked

Field	Description
EVAL_TYPE_DESC	Evaluation type (e.g. FCE/PCE variants, Investigation, Self-Disclosure). <i>Data:</i> includes a “Self-Disclosure” category
EVAL_LEAD_AGENCY	Lead agency for the evaluation (EPA / State / Local)
EVAL_DATE	Date of the evaluation

Violation block

Field	Description
VIOL_FLAG	Flag indicating a violation on the row
VIOL_SORT_ORDER	Internal ordering of the violation
FOUND_VIOLATION	<i>Data:</i> Y for all 66,655 rows
VIOL_ACTIVITY_ID	Violation activity identifier. <i>Data + EPA:</i> some rows carry system-generated IDs (prefixes 9906/9913) that, per EPA, “did not have an actual violation activity identification number” and were “system generated for purposes of creating the pipeline table”
VIOL_TYPE	Violation type. <i>Data:</i> HPV, FRV, plus placeholder values (blank, and “Linked to Viol. Below”) on the system-generated rows
VIOL_TYPE_SORT	Internal ordering by violation type
VIOL_LEAD_AGENCY	Lead agency for the violation (EPA / State / Local; often blank on placeholder rows)
VIOL_PROGRAMS	Regulatory program(s) violated
VIOL_POLLUTANT_CODES / VIOL_POLLUTANT_DESCS	Pollutant code(s) and description(s)
VIOL_START_DATE	Violation start date
VIOL_END_DATE	Violation end/resolution date (text field, mixed format)
VIOL_END_DATE_DATE	Parsed date version of VIOL_END_DATE

Enforcement-action block

Field	Description
EA_FLAG	Flag indicating an enforcement action on the row
EA_SORT_ORDER	Internal ordering of the enforcement action
EA_ACTIVITY_ID	Enforcement-action activity identifier
EA_FEA_ACTIVITY_ID	Separate identifier for the formal enforcement action
EA_TYPE	Enforcement-action type (e.g. Notice of Violation, Administrative – Formal, Judicial)
EA_DATE	Enforcement-action date
FEA_ISSUE_DATE_FLAG	Flag for the formal-enforcement-action issue date
EA_PENALTY_AMT	Monetary penalty. <i>Data:</i> non-zero on 14,427 rows
EA_COMP_ACTION_COST	Supplemental compliance / environmental-project cost. <i>Data:</i> non-zero on 94 rows

Lead-agency note (*data*). There is no enforcement-action-level lead-agency field; the only lead-agency columns are EVAL_LEAD_AGENCY and VIOL_LEAD_AGENCY. Of the 94 non-zero EA_COMP_ACTION_COST

rows, 83 have a blank VIOL_LEAD_AGENCY, so agency attribution via that field is incomplete.